SOGAT meets Java

Razor

```java
public class Razor {

    private Razor() { }

    public static final boolean TRUE = true;

    public static final boolean FALSE = false;

    public static <A> A ite(boolean b, A x, A y) {
        return b ? x : y;
    }

    public static int plus(int x, int y) {
        return x + y;
    }

    public static boolean isZero(int x) {
        return x == 0;
    }

}
```

System T

```java
@FunctionalInterface
public interface Recursor<A> {

    A apply(int x, A p);

}
```

```java
import java.util.function.Function;

public class SystemT {

    private SystemT() { }

    public static final int ZERO = 0;

    public static int suc(int x) {
        return x + 1;
    }

    public static <A> A ite(A z, Function<A, A> s, int x) {
        return x == ZERO ? z : s.apply(ite (z, s, x - 1));
    }

    public static <A> A rec(A z, Recursor<A> s, int x) {
        return x == ZERO ? z : s.apply(x - 1, rec(z, s, x - 1));
    }

    public static int pred(int x) {
        return rec(ZERO, (y, z) -> y, x);
    }

}
```